

FINLEY JOHN NEILSON

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SUMMARY

Full-stack developer and Honours Computer Science student at the University of Auckland, designing React and Next.js interfaces on PostgreSQL backends. Researching real-time 3D graphics and shaders, with a co-authored paper accepted to OzCHI 2026.

EDUCATION

University of Auckland

Bachelor of Science (Honours), Computer Science | GPA: 8.7 / 9

March 2026 – November 2026 (expected)

Auckland, NZ

Victoria University of Wellington

Bachelor of Science, Computer Science and Philosophy | GPA: 7.66 / 9

February 2021 – November 2025

Wellington, NZ

EXPERIENCE

Software Developer

WDCC – Web Development & Consulting Club

April 2026 – Present

Auckland, NZ

- Collaborating in an Agile environment to build the official Linux User Group website.
- Contributing through weekly team meetings and delivering assigned tickets throughout the year.

Computer Science Tutor

COMP 112, Victoria University of Wellington

February 2022 – July 2022

Wellington, NZ

- Provided tutoring support to undergraduate students by marking assignments and helping people in labs.
- Received positive feedback for effective teaching methods and clear communication.

Junior Chef

Arcimboldi

October 2021 – October 2024

Seatoun, Wellington

PROJECTS

GeoScratch (Honours Research) | React, Three.js, Blockly, WebGL, Supabase/PostgreSQL, HCI

[GitHub](#)

- Co-authored *GeoScratch – A Block Based Language for Geometry Education* on the earlier GeoScratch system, accepted to OzCHI 2026 (Australian Conference on Human-Computer Interaction).
- Building a block-based 3D environment for learning geometry and linear algebra.
- Implementing perceptual rendering for depth disambiguation in dense scenes: a GPU halo pass that opens depth-ordered gaps where lines and vectors cross.
- Reducing visual clutter with zoom-invariant glyph sizing and a force-based declutter layer for 3D labels.
- Designing animation as explanation rather than decoration, revealing a construction in the order its blocks build it.
- Instrumenting the tool for user studies: a Supabase backend with anonymous participant accounts, cohort tagging and per-attempt logging behind row-level security.

Procedural Terrain Generation | C++, OpenGL, GLSL, ImGui

[GitHub](#)

- Developed a procedural terrain generation system using fractal Perlin noise functions.
- Implemented a thermal erosion simulation to improve terrain realism through iterative height redistribution.
- Built a shader-based terrain texturing system blending sand, grass, stone, and snow textures based on elevation.

Linux User Group Website | Next.js, TypeScript, Tailwind CSS, Docker

[GitHub](#)

- Developing a modern, responsive web application for the LUG community using Next.js and TypeScript.
- Implementing UI components with Tailwind CSS, ensuring a consistent design language across the platform.
- Built a registration form that remains fully functional with JavaScript disabled, per client requirements.

Shape Transform | React, TypeScript, Three.js, Node.js, Express, MongoDB

[GitHub](#)

- Built a full-stack 3D spatial-reasoning puzzle game with Three.js rotation gameplay, timers, and a leaderboard.
- Implemented a REST API with JWT authentication and MongoDB for accounts and score tracking, and backend tests.

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, HTML, CSS, Go, C, C++ | **Web:** MERN Stack (MongoDB, Express, React, Node.js), Next.js, Tailwind CSS | **Graphics:** Three.js, WebGL, OpenGL, GLSL | **Tools:** Git, Docker, Neovim, WSL, JetBrains IDEs | **Databases:** PostgreSQL, MongoDB